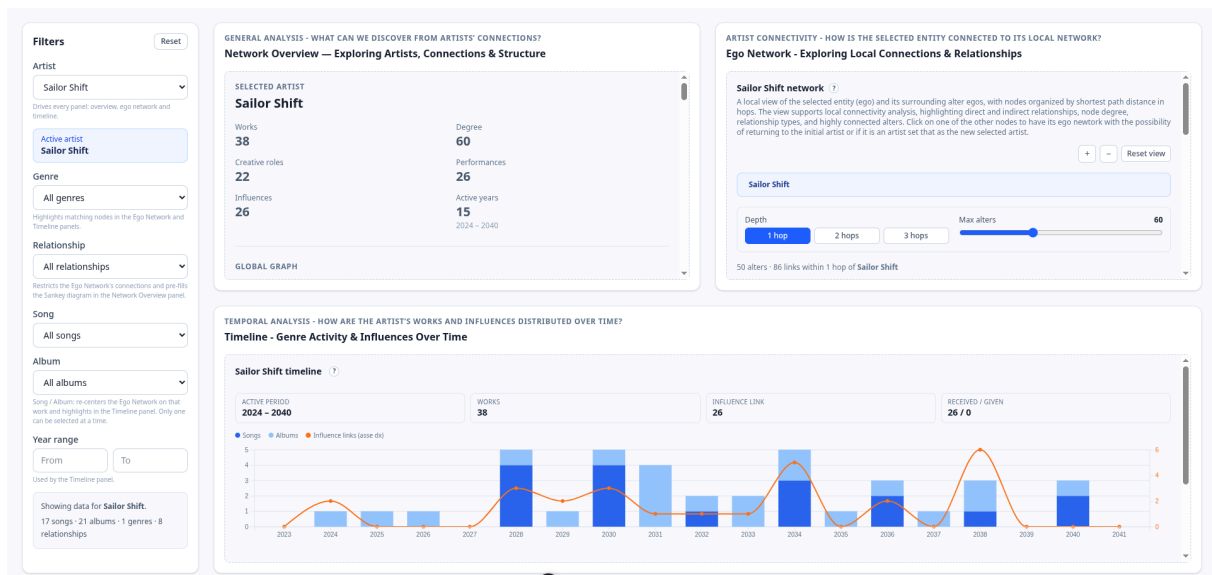


University of Pisa
Department of Computer Science
Data Science and Business Informatics

Designing Interactive Visual Analytics for Knowledge Graph Exploration

Visual Encodings, Interaction, and User Workflow
for the VAST 2025 Design Challenge

Aurora Bracciotti
Visual Analytics — 602AA
VAST 2025 Design Challenge



Academic Year 2025/2026

Contents

1	Introduction	2
2	Dataset and Data Processing	2
3	Design Goals	2
3.1	Relationship Discovery	3
3.2	Global and Local Exploration	3
3.3	Temporal Exploration	3
3.4	Supporting Different Levels of Expertise	3
4	Dashboard Design	3
4.1	Filter Panel	3
4.2	Overview Network	4
4.3	Ego Network	5
4.4	Artist Timeline	6
5	Interaction and User Support	6
5.1	Coordinated Filtering	6
5.2	Selection and Navigation	7
5.3	Explanatory Support	7
6	Analytical Scenario	7
7	Design Process	8
8	Limitations	8
9	Conclusion	9

1 Introduction

This project was developed in response to the *VAST 2025 Design Challenge*, which investigates how visual analytics can facilitate the exploration of knowledge graphs. The selected analytical goal is the discovery of new information and relationships. The design takes inspiration from *Mini-Challenge 1*, whose music knowledge graph provides a suitable context for investigating an artist’s career, collaborations, and influences.

Knowledge graphs provide a way to represent complex domains through entities and the relationships connecting them. In the context of music, a single graph can describe artists, songs, albums, genres, and different forms of collaboration and influence. However, the number and variety of these relationships can make it difficult to understand an artist’s position within the broader network and identify meaningful connections.

The proposed solution is an interactive dashboard designed to support the exploration of an artist from multiple perspectives. An *Overview Network* provides the broader context of the graph, while an *Ego Network* enables a more focused investigation of the connections surrounding a selected artist. A *Timeline* complements the network views by placing the artist’s works and influences in a temporal context. Interactive filters and coordinated views allow users to progressively narrow their exploration, moving from a general overview to more detailed investigations. The dashboard is intended to make the exploration of the knowledge graph accessible to users with different levels of expertise, allowing them to investigate an artist’s relationships and career without requiring specialized knowledge of graph analysis.

2 Dataset and Data Processing

The project uses the music knowledge graph provided for the *VAST 2025 Mini-Challenge 1*. The dataset represents a heterogeneous network of music-related entities, including artists, songs, albums, genres, musical groups, and other entities involved in the music industry. The connections between these entities are represented through typed relationships describing different forms of interaction, such as collaborations, influences, performances, and other music-related relations.

Dataset	Nodes	Links	Description
VAST 2025 MC1	17,412	37,857	Heterogeneous music knowledge graph

Table 1: Structural characteristics of the VAST 2025 Mini-Challenge 1 dataset.

The graph is used as the basis for the different views of the dashboard. Its heterogeneous structure makes it possible to investigate an artist both in relation to the broader network and through the specific entities and relationships surrounding them. In particular, artist relationships, works, and influences provide the information required to construct the *Overview Network*, *Ego Network*, and *Artist Timeline* views.

Before being used by the visualizations, the original data is processed to obtain a consistent representation of the graph. The application first loads the dataset (`graphLoader.vue`) and then applies a series of transformations to normalize the data (`graphTransforms.vue`) and prepare it for visualization. Additional information is also derived from the graph to support the metrics (`metrics.vue`) and relationship-based analysis (`relationships.vue`) used throughout the dashboard.

3 Design Goals

The design was guided by the objective of supporting the discovery of new information and relationships within the knowledge graph. Rather than presenting all available information in

a single visualization, the dashboard was designed to provide complementary perspectives that users can explore according to their analytical needs.

3.1 Relationship Discovery

The primary goal is to help users identify relevant relationships surrounding an artist. The interface provides information about the distribution of relationships in the broader graph and then allows users to investigate the specific connections of an individual artist.

3.2 Global and Local Exploration

A second goal is to connect the global structure of the knowledge graph with the local context of a selected artist. The Overview Network provides information about the overall network, while the Ego Network allows users to focus on the neighborhood of a specific entity.

3.3 Temporal Exploration

The design also considers the temporal dimension of an artist’s career. The Artist Timeline allows users to relate works and influence relationships to specific years, providing a complementary perspective that cannot be obtained from the network structure alone.

3.4 Supporting Different Levels of Expertise

The dashboard is designed for users with different levels of familiarity with graph-based data analysis. Visual summaries provide an initial interpretation of the data, while explanatory elements and interaction guidance help users understand more complex visualizations and controls.

4 Dashboard Design

The dashboard is organized as a fixed workspace with three main visualization panels. The *Overview Network* and *Ego Network* are positioned next to each other in the upper part of the interface, while the *Artist Timeline* occupies the entire width of the lower section.

The dashboard does not require vertical scrolling. This fixed arrangement was chosen to keep the main views visible simultaneously and to support comparison between the different perspectives. The separation into multiple panels also avoids concentrating all available information in a single visualization, which could make the interface unnecessarily complex for non-expert users while showing irrelevant information to users who want to focus on a specific aspect of the data.

The left side of the dashboard contains the filtering system, which remains visible throughout the exploration. The artist selection acts as the main contextual filter: once an artist is selected, the available values in the other filters are constrained to information relevant to that artist. The available filters include artist, genre, song, album, year range, and relationship type.

The dashboard follows a card-based organization. Each panel can be opened or closed independently, allowing users to control the amount of information displayed. Cards can also be closed directly from their current position, so users do not need to return to the beginning of the dashboard to manage the interface.

4.1 Filter Panel

The filter panel provides the main mechanism for defining the analytical context. Users can select an artist, genre, song, album, year range, or relationship type. The filters are coordinated so that selections remain consistent with the currently selected artist.

A short description is provided below each filter to indicate which parts of the dashboard are affected by the corresponding selection. This makes the effect of the interaction explicit and helps users understand how their choices will modify the visualizations.

The bottom of the filter panel contains a numerical summary of the selected artist, including information about their works, genres, and relationships. A reset control allows users to remove the active filters and return to the initial state.

4.2 Overview Network

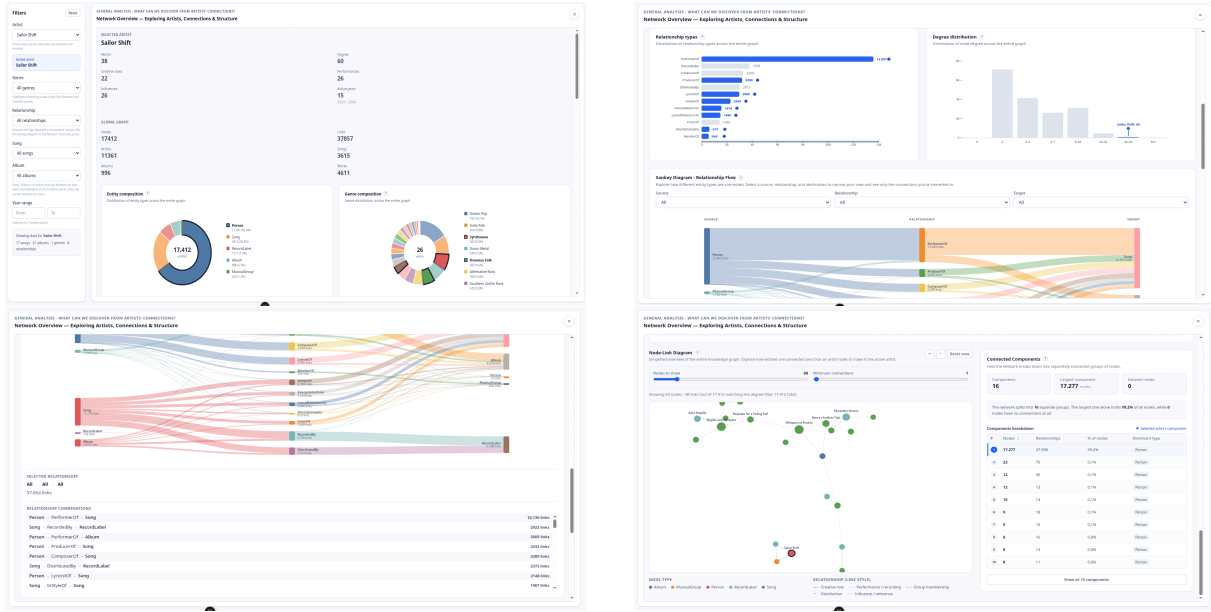


Figure 1: Overview Network panel.

The Overview Network provides a general view of the graph while contextualizing the analysis around the selected artist.

The panel begins with a summary of the selected artist, providing an initial overview before the user examines the visualizations. This is followed by a representation of the complete network and a compact 2-by-2 grid containing four summaries: entity composition, genre distribution, relationship distribution, and degree.

The relationship distribution is represented using a horizontal bar chart. The bars show the frequency of the different relationship types in the graph, while the selected artist is highlighted within the corresponding relationship information. This allows users to compare the artist's relationships with the overall distribution without requiring a separate visualization.

The entity and genre distribution are represented as a pie chart, with the relevant artist-related information highlighted using a distinct border. The degree visualization provides information about the connectivity of artists, allowing users to compare the selected artist with other entities in the graph.

The panel also contains a Sankey diagram that represents the structure of relationships through three levels:

$$\text{Source Entity} \rightarrow \text{Relationship} \rightarrow \text{Target Entity}$$

This representation allows users to understand not only which relationships are present, but also which types of entities are connected through them.

The lower part of the Overview Network contains a simplified node-link diagram together with information about the connected components of the graph. The node-link diagram provides a structural representation of the graph, while limiting the number of displayed nodes to maintain readability. The selected artist is highlighted so that its position can be identified within the network.

The node-link visualization supports several interactions. Users can hover over a node to highlight its connected entities, drag nodes to explore the structure, and zoom and pan to navigate the graph. Clicking an artist selects it and allows the user to investigate its individual network in the Ego Network panel. Two controls further regulate the visualization: *Maximum nodes* limits the number of displayed nodes, while *Minimum connections* hides nodes with fewer connections than the selected threshold. A reset control restores the original zoom and position.

The connected-components section complements the node-link representation with a more abstract description of network connectivity. It reports the number of connected components, isolated nodes, and the largest component, together with the number and percentage of nodes belonging to the different components. The entity composition of the components is also provided, allowing users to understand what types of entities form the different parts of the graph.

4.3 Ego Network

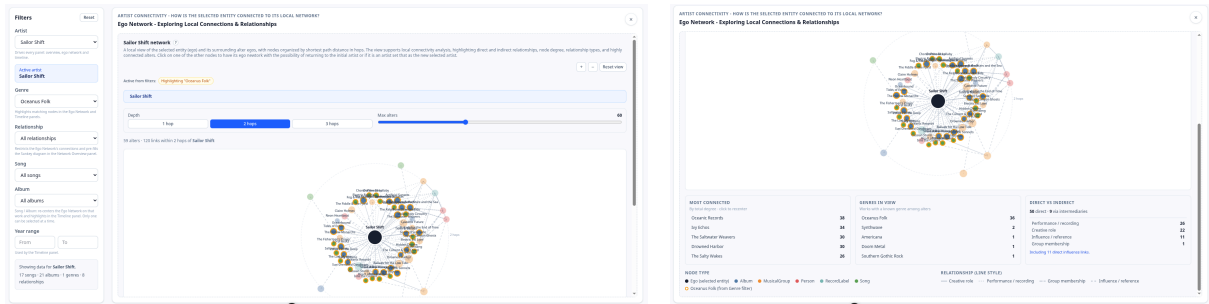


Figure 2: Examples of the interactive elements of the dashboard.

The Ego Network provides a more focused representation of the selected artist and their surrounding relationships. While the Overview Network describes the broader graph, the Ego Network reduces the amount of information displayed and allows the user to investigate the local neighborhood of an entity.

Users can explore the network up to three hops from the selected artist and control the number of alters displayed. This makes it possible to adjust the level of detail according to the complexity of the artist's network.

The visualization uses consistent legends and visual encodings with the Overview Network. Maintaining the same representation across panels reduces the need for users to learn new visual conventions when moving from the global to the local view.

A summary is displayed alongside the ego network, providing information about the connected entities, the genres represented in the neighborhood, and the direct and indirect relationships. This summary provides a compact description of the selected local network before the user investigates individual connections.

The Ego Network is interactive. Users can zoom and navigate the graph, select entities, and change the entity being investigated. A dedicated control allows users to return to the initially selected artist, preserving the original analytical context after navigating through other entities.

The Ego Network is also affected by the global filters. For example, selecting a genre or relationship type changes the information displayed in the network, allowing users to investigate a specific subset of the artist's connections.

4.4 Artist Timeline

The Artist Timeline introduces a temporal perspective on the selected artist’s career. The main visualization represents the number of works and influence relationships for each year. Songs and albums are visually distinguished, while influence relationships provide information about the artist’s incoming and outgoing connections.

Selecting a year allows users to inspect the corresponding information in greater detail. The selected period can include songs, albums, incoming influences, outgoing influences, and the names and roles of the related entities.

Additional information identifies the entities that most strongly influenced the selected artist and the entities that contributed most to the influence within the artist’s genre. This allows the user to move from the temporal distribution of relationships to the specific entities involved.

The timeline begins with a summary of the artist’s period of activity, number of works, and influence relationships. This provides an initial context for interpreting the temporal visualization. The activity period also helps users choose meaningful year ranges. If the selected range does not contain any works by the artist, the interface explicitly indicates that no timeline is available for that period.

The currently selected year range is displayed within the panel. Leaving the year range unspecified allows the user to examine the complete available period of the artist’s career.

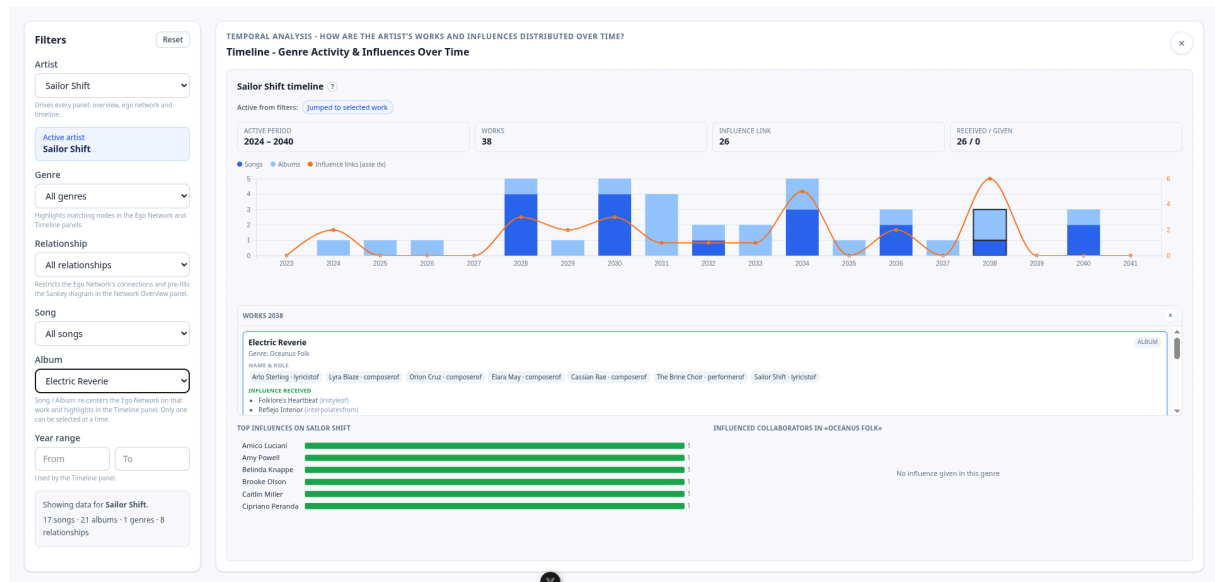


Figure 3: Timeline panel showing the selected artist’s activity with the album filter applied.

5 Interaction and User Support

The dashboard relies on coordinated interaction to allow users to move between different levels of analysis without losing their context.

5.1 Coordinated Filtering

The filters act as a shared mechanism across the dashboard. Selecting an artist defines the main analytical context and constrains the available values in the remaining filters. Subsequent selections can therefore be used to investigate particular genres, songs, albums, relationship types, or periods associated with that artist.

The effects of filters are reflected dynamically in the different panels. The information below each filter explicitly indicates which visualization is affected, helping users understand the consequences of their selections.

When filters are active, the Ego Network and Timeline display a reminder of the current filtering state. This provides persistent contextual information and makes it easier for users to identify and remove filters that are no longer relevant.

5.2 Selection and Navigation

Selection is used to connect different levels of analysis. In particular, selecting an artist in the Overview Network can make that artist the focus of the Ego Network, allowing users to move from the global graph to a local investigation.

Within the Ego Network, users can continue exploring by selecting other entities. Because this can change the current focus, a dedicated control allows users to return to the originally selected artist.

Zooming, panning, dragging, and hovering are mainly used to support the exploration of the node-link representation. These interactions allow users to inspect dense areas of the graph without requiring the entire network to be displayed at maximum detail.

5.3 Explanatory Support

An important aspect of the design is the integration of explanations directly into the interface. Each card provides a short description of its content, while the title is accompanied by a more general analytical question.

This distinction provides two levels of interpretation. A general question describes what the user can investigate, making the purpose of the visualization more accessible to non-expert users. The more specific visualization title provides a concise technical description for users who are already familiar with the concepts.

Individual visualizations also provide a “?” control containing additional information. The explanation describes what the visualization represents and, where necessary, how to interact with it. More complex visualizations, such as the Sankey diagram and node-link diagram, include specific instructions for their controls and interactions.

This explanatory layer is intended to reduce the amount of prior knowledge required to use the dashboard while avoiding the need to provide extensive instructions permanently on the main interface.

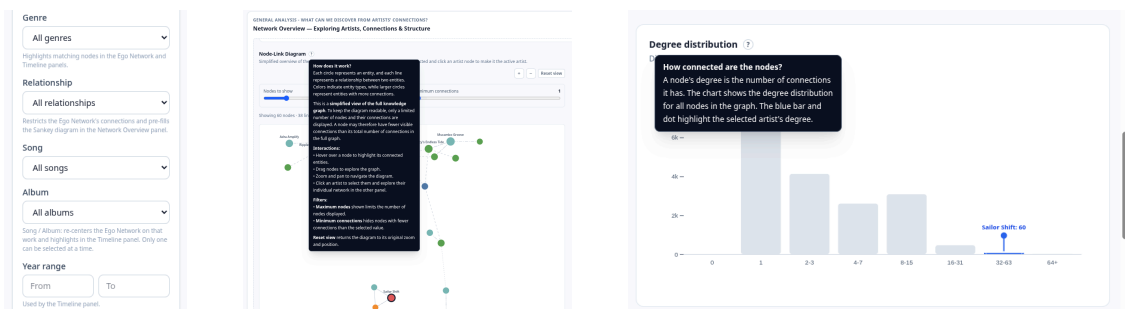


Figure 4: Examples of the interactive elements of the dashboard.

6 Analytical Scenario

The dashboard can be used to investigate an artist’s career and discover relationships within the music knowledge graph. A possible exploration can start by selecting an artist such as Sailor

Shift from the artist filter.

The Overview Network provides the first level of analysis. The artist summary and network statistics give an initial understanding of the selected entity, while the relationship and degree distributions provide a comparison with the broader graph. The Sankey diagram can then be used to investigate which entity types are connected through which relationship types.

After identifying a potentially relevant relationship, the user can move to the Ego Network to investigate the artist’s local neighborhood. The number of hops and alters can be adjusted to control the complexity of the view. Direct and indirect connections can then be explored by selecting individual entities.

The Timeline can subsequently be used to determine when relevant works or influence relationships occurred. Selecting a specific year reveals the entities associated with that period and allows the user to relate network connections to the development of the artist’s career.

At any point, filters can be used to narrow the investigation. For example, the user can restrict the exploration to a specific genre or relationship type, or focus on a particular song or album. Because the views are coordinated, these selections are reflected across the relevant visualizations.

This workflow supports a progression from broad network exploration to local relationship inspection and finally to temporal analysis. The combination of these perspectives can help users identify connections that may not be immediately apparent when examining the complete knowledge graph.

7 Design Process

The design process was primarily driven by the analytical task rather than by the availability of individual visualization techniques. The starting point was the question of how a user could discover new relationships around an artist and what information would be required to support this exploration.

For each part of the dashboard, the design process focused on identifying the analytical question that the visualization should answer. This helped avoid adding visualizations simply because additional data was available. Instead, each view was selected according to the type of information it could contribute to the overall exploration.

The Overview Network was designed to provide the initial context and describe the structure of the broader graph. The Ego Network was introduced to reduce the complexity of the graph and focus the analysis on the relationships surrounding one artist. The Timeline was then used to introduce the temporal dimension and connect relationships with the development of the artist’s career.

Another important design consideration was the balance between information richness and readability. Combining every visualization into a single large view could provide more information but would also increase visual complexity. Dividing the interface into separate panels allows users to focus on the aspect of the analysis that is relevant to their current question.

The design process also considered users with different levels of expertise. Rather than assuming that users would already understand the purpose of every visualization, analytical questions, descriptions, filter explanations, and interaction instructions were incorporated into the interface. These elements provide guidance when needed while allowing experienced users to move directly to the visualizations.

8 Limitations

The proposed dashboard is primarily suited to heterogeneous knowledge graphs containing typed entities and relationships. It is particularly appropriate for domains where entities can be explored from both structural and temporal perspectives.

A first limitation concerns scalability. The complete graph contains 17,412 nodes and 37,857 links, making a full node-link representation difficult to inspect at once. The Overview Network therefore uses a simplified representation, while the Ego Network restricts the analysis to a limited number of hops and alters. Highly connected entities may nevertheless produce complex local networks.

The design also depends on the availability of temporal information. The Timeline requires works and relationships to be associated with years in order to provide a meaningful temporal representation. Knowledge graphs without sufficient temporal information would therefore not benefit from this view to the same extent.

The filtering system relies on consistent entity and relationship types. If relationship labels or entity classifications were ambiguous or inconsistent, the effectiveness of the filtering and summary mechanisms would be reduced.

The dashboard is designed primarily for desktop-sized displays. Its fixed layout and simultaneous presentation of multiple views are intended for a relatively large screen and may require adaptation for smaller devices.

9 Conclusion

This project presents an interactive visual analytics dashboard for exploring relationships within a heterogeneous music knowledge graph. Inspired by the *VAST 2025 Design Challenge* and *Mini-Challenge 1*, the design focuses on the discovery of new information and relationships surrounding an artist.

The dashboard combines an *Overview Network*, an *Ego Network*, and an *Artist Timeline*, supported by coordinated filters and interactive exploration. Approaching the challenge from an open-ended design perspective influenced the project by making the definition of the analytical focus an important part of the design process. Rather than starting with a fixed visualization or a predetermined set of charts, the design was organized around the questions that users should be able to answer when exploring the knowledge graph.

One of the main challenges was deciding which information should be presented and at what level of detail. The knowledge graph contains many different types of entities and relationships, but displaying all of them simultaneously would make the analysis more difficult. This led to the decision to distribute the information across complementary views, each addressing a specific aspect of the exploration.

Another challenge was designing an interface that could accommodate different levels of expertise. The addition of analytical questions, contextual descriptions, filter explanations, and interaction guidance was intended to make the dashboard easier to approach while preserving detailed information for users who want to perform a more focused analysis.

The final design therefore reflects a balance between exploration and guidance. Instead of simplifying the underlying graph by removing its complexity entirely, the dashboard provides different ways of accessing that complexity depending on the user’s current analytical needs.

Acknowledgement

OpenAI ChatGPT and Anthropic Claude were used as implementation assistants for code organization, debugging, testing, refinement, and report preparation. The data-processing components were adapted from an existing implementation found in a public fork of the project repository and revised to fit the proposed visualization system. The analytical goals, design choices, interpretation, and final decisions remain the author’s own.